

Revision Notes for Class 12 Chemistry

Chapter 1 – Solutions

1.1 Definition

A solution is a mixture of two or more components that is homogeneous.

Take, for instance, ordinary salt in water.

1.2 Classification

Binary solutions are solutions that have two components in them. Components are substances that are used to make a solution.

The solvent is the component that is present in the greatest quantity. The physical condition of a solution is determined by the solvent.

Solute refers to the other component in the solution that is present in smaller amounts.

Each element could be solid, liquid, or gaseous.

Types of Solution	Solute	Solvent	Common Examples
Gaseous Solutions	Gas	Gas	Oxygen and nitrogen gas mixture
	Liquid	Gas	a mixture of chloroform and nitrogen gas
	Solid	Gas	nitrogen gas with camphor

Liquid Solutions	Gas	Liquid	Water with oxygen dissolved in it
	Liquid	Liquid	Ethanol is a liquid that can be dissolved in water.
	Solid	Liquid	Glucose that has been dissolved in water
Solid Solutions	Gas	Solid	Hydrogen solution in palladium
	Liquid	Solid	Mercury and sodium amalgam
	Solid	Solid	dissolving copper in gold

1.3 Strength of Solutions

Strength of solution refers to the amount of solute dissolved per unit of solution or solvent. There are several methods for determining the strength of a solution.

1. Mass Percentage (%w/w): "It denotes the mass of a component in 100 g of solution."

$$\text{Mass \% of a component} = \frac{\text{Mass of component in the sol.}}{\text{Total Mass of sol.}} \times 100$$

2. Volume Percentage (%v/v): "It represents the volume of a component in a solution of 100mL."

$$\text{Volume \% of a component} = \frac{\text{Volume of component}}{\text{Total Volume of sol.}} \times 100$$

3. Mass by Volume Percentage (% w / v): "It denotes the mass of the solute in grams in 100mL of solution."

$$\text{Mass by vol, percent} = \frac{\text{Mass of solute in g}}{\text{Vol. of sol. in mL}} \times 100$$

4. Parts Per Million (ppm):

$$\text{Parts per million} = \frac{\text{No. of parts of the component}}{\text{Total no. of all the components of sol.}} \times 10^6$$

Mass to mass, volume to volume, and mass to volume are all ways to express concentration in parts per million.

5. Mole Fraction (x): "It denotes the number of moles of a solute in one mole of solution."

$$\text{Mole fraction} = \frac{\text{No. of moles of the component}}{\text{Total no. of moles all the components}}$$

If the amount of moles of A and B in a binary mixture is n_A and n_B , respectively, the mole fraction of A will be

$$x_A = \frac{n_A}{n_A + n_B}$$

6. Molarity, M: "It represents the number of moles of solute in 1 litre of solution."

$$\text{Molarity, M} = \frac{\text{Moles of solute}}{\text{Vol. of sol in L}}$$

Molarity is measured in milligrams per litre and is denoted by the letters ' M ' or 'Molar.' "The density of a solution is its mass per unit volume."

$$\text{Density, d} = \frac{\text{Mass of sol.}}{\text{Vol. of sol.}} = m / V$$

7. Molality, m: “It denotes the number of moles of solute present per kilogramme of solvent.”

$$\text{Molality, } m = \frac{\text{Moles of solute}}{\text{Mass of solvent in kg}}$$

Molality is measured in mol/kg, which can also be written as 'm' or 'molal.'

8. Normality, N: It denotes the number of solute equivalents in 1 litre of solution.

$$\text{Normality, } N = \frac{\text{No. of Equivalents of solute}}{\text{Vol. of sol. in L}}$$

$$\text{No. of equivalents, } eq = \frac{\text{Weight}}{\text{Equivalent weight (W / E)}}$$

$$E = \frac{M}{Z} \text{ (Z is the valency factor).}$$

Some Important Relationships

Dilution Law: When we dilute a solution with solvent, the amount of solute remains constant, and we can write:

$$M_1 V_1 = M_2 V_2 \text{ and } N_1 V_1 = N_2 V_2$$

Molarity and Normality

$$\text{Normality} = z \times \text{Molarity}$$

Important:

The temperature has no effect on the mass per cent, ppm, mole fraction, or molality, however temperature influences molarity and normality. This is because the volume is affected by temperature whereas mass is not.

2. Vapour Pressure

2.1 Definition

At a given temperature, the vapour pressure of a liquid/solution is the pressure exerted by vapours in equilibrium with the liquid/solution.

Vapour pressure $\propto$ escaping tendency

2.2. Vapour Pressure of Liquid Solutions and Raoult's Law :

(Raoult's law for volatile solutes)

Raoult's law states that the partial vapour pressure of each component in a solution of volatile liquids is directly proportional to its mole fraction.

$$p_1 \propto x_1$$

and $p_1 = p_1^0 x_1$

Component 2 is the same way.

$$p_2 = p_2^0 x_2$$

The total pressure $P_{(\text{total})}$ over the solution phase in the container, according to Dalton's partial pressure law, is equal to the sum of the partial pressures of the solution's components and is given as:

$$P_{(\text{total})} = p_1 + p_2$$

We get by substituting the values of p_1 and p_2

$$P_{(\text{total})} = x_1 p_1^0 + x_2 p_2^0$$

$$= (1 - x_2) p_1^0 + x_2 p_2^0$$

$$= p_1^0 + (p_2^0 - p_1^0) x_2$$

At constant temperature, the plot of vapour pressure and mole fraction of an ideal solution. The partial pressures of the components are shown by the dashed lines I and II. p_1 and p_2 are directly proportional to x_1 and x_2 , respectively, as can be seen from the plot. The total vapour pressure is represented in the figure by line III.

Mole Fraction in The Vapour Phase

Using Dalton's rule of partial pressures, if y_1 and y_2 are the mole fractions of components 1 and 2 in the vapour phase, then:

$$p_1 = y_1 p_{\text{total}}$$

$$p_2 = y_2 p_{\text{total}}$$

Generally

$$p_i = y_i p_{\text{total}}$$

2.3. Vapour Pressures of Solutions of Solids in Liquids and Raoult's Law

(Raoult's law for non-volatile solutes)

When a non-volatile solute is added to a solvent to make a solution, the number of solvent molecules leaving from the surface is reduced, lowering the vapour pressure.

The amount of non-volatile solute present in the solution, regardless of its composition, determines the drop in solvent vapour pressure.

In its most general form, Raoult's law states that the partial vapour pressure of each volatile component in a solution is directly proportional to its mole fraction for every solution.

$$p_1 \propto x_1$$

$$p_1 = x_1 p_1^0 = p_{\text{total}}$$

If Raoult's law holds true for all concentrations, a solution's vapour pressure will vary linearly from zero to the pure solvent's vapour pressure.

2.4. Ideal and Non-Ideal Solutions

Ideal Solutions:

An ideal solution is one in which each component follows Raoult's rule under all temperature and concentration circumstances.

Properties of Ideal Solutions:

$$\Delta H_{\text{MIXING}} = 0$$

$$\Delta V_{\text{MIXING}} = 0$$

The attractive forces between the A-A and B-B molecules are approximately comparable to those between the A-B molecules.

For example, benzene and toluene solution, n-hexane and n-heptane solution

Non – Ideal Solutions:

When a solution deviates from Raoult's law over a wide concentration range, it is referred to as a non-ideal solution.

Solutions Showing Positive Deviation from Raoult's Law:

- Solute-Solute(B-B) and Solvent-Solvent(A-A) forces are weaker than Solvent-Solute(A-B) forces.
- The vapour pressure is higher than the law predicts.

$$\Delta H_{\text{MIXING}} > 0$$

$$\Delta V_{\text{MIXING}} > 0$$

Ethanol with acetone, for example, or carbon disulphide and acetone.

Negative Deviations from Raoult's Law are Shown in the Following Solutions:

- The Solvent-Solute(A-B) force is more powerful than the other two.
- The vapour pressure is lower than the law predicts.

$$\Delta H_{\text{MIXING}} < 0$$

$$\Delta V_{\text{MIXING}} < 0$$

2.5. Azeotropes

Azeotropes are binary mixes that boil at the same temperature and have the same composition in the liquid and vapour phases.

Minimum Boiling Azeotrope

Minimum boiling azeotrope at a certain composition is formed by solutions that demonstrate a considerable positive departure from Raoult's law.

For example, an azeotrope with a boiling point of 351.15 K is formed by a mixture of ethanol and water containing roughly 95% ethanol.

The temperature at which water boils - composition Large positive deviations are depicted in this diagram for solutions (Minimum boiling azeotrope).

Maximum Boiling Azeotrope:

Maximum boiling azeotrope at a certain composition is formed by solutions that demonstrate a substantial negative divergence from Raoult's law. A mixture of 68 per cent nitric acid and water produces an azeotrope with a boiling point of 393.5 K.

The temperature at which water boils - composition Large negative variances are depicted in this diagram for solutions. (Azeotrope with the highest boiling point)

3. Solubility

3.1. Solubility of a Solid in Liquid

A substance's solubility refers to how much of it can be dissolved in each amount of solvent.

Factors Impacting a Solid's Solubility in a Liquid Include:

1. Nature of solute and solvent: Like disintegrates into like. Sodium chloride and sugar, for example, dissolve quickly in water, but naphthalene and anthracene do not. Naphthalene and

anthracene, on the other hand, dissolve quickly in benzene, but sodium chloride and sugar do not.

2. Temperature: If $(\Delta_{\text{sol}}H^0)$ the solubility of a nearly saturated solution increases as the temperature rises, and if $(\Delta_{\text{sol}}H^0)$, the solubility falls as the temperature rises.

3. Pressure's Effect: Solids and liquids are highly incompressible, so, this has little effect.

3.2. Henry's Law

The solubility of a gas in a liquid is directly proportional to the pressure of the gas at a fixed temperature, according to Henry's law.

“The partial pressure of the gas in the vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution,” according to the most frequent version of Henry's law. This is written as: The constant K_H stands for Henry's law.

K_H Characteristics

The type of the gas determines K_H . The lower the solubility of the gas in the liquid, the higher the value of K_H at a given pressure. K_H values increase as temperature decreases, indicating that gas solubility increases as temperature decreases.

Henry's Law has a Variety of Applications.

- In the manufacture of carbonated drinks.
- During a deep-water dive.
- Climbers and those who live at high altitudes will benefit from this.

Henry's Law as a Specific Example of Raoult's Law

Raoult's law states that $p_i = x_i p_i^0$

One of the components of a gas in a liquid solution is so volatile that it exists as a gas. According to Henry's law, it is soluble in water.

$$p_i = K_H x$$

As a result, Raoult's law becomes a specific case of Henry's law, in which K_H equals p_1^0 .

4. Colligative Properties

Colligative qualities are those that are dependent on the number of solute particles present in a solution, regardless of their type relative to the total number of particles present.

There are four qualities that are colligative:

1. A Relative Decrease in Vapor Pressure
2. Boiling Point Elevation
3. Freezing point depression
4. Osmotic pressure is the factor to consider.

4.1. Relative Lowering of Vapour Pressure

When a non-volatile solute is added to a solvent, the vapour pressure decreases. The lowering of vapour pressure with respect to, the vapour pressure of the pure solvent is called “Relative lowering in vapour pressure”.

Raoult's Law States That:

$$p_1 = x_1 p_1^0$$

The reduction in the vapour pressure of solvent (Δp_1) is given as:

$$\Delta p_1 = p_1^0 - p_1 = p_1^0 - p_1^0 x_1 = p_1^0 (1 - x_1)$$

Knowing that $x_2 = 1 - x_1$ equation then reduces to $\Delta p = x_2 p_1^0$

This equation can be written as

$$\frac{\Delta p}{p_1^0} = \frac{p_1^0 - p_1}{p_1^0} = x_2$$

As previously stated, the expression on the left-hand side of the equation is known as relative lowering of vapour pressure and equals the mole fraction of the solute. The equation above can be represented as:

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1 + n_2} \left(\text{since } x_2 = \frac{n_2}{n_1 + n_2} \right)$$

The number of moles of solvent and solute present in the solution are denoted by n_1 and n_2 , respectively. Because $n_2 \ll n_1$ for dilute solutions, we can ignore n_2 in the denominator.

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1}$$

Or

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times M_1}{M_2 \times w_1}$$

The masses w_1 and w_2 are the masses of the solvent and solute, respectively, while M_1 and M_2 are the molar masses of the solvent and solute.

4.2. Elevation in Boiling Point

The temperature at which a liquid's vapour pressure equals atmospheric pressure is known as the boiling point of the liquid.

The vapour pressure of the solvent decreases when a non-volatile solute is added, hence the needed temperature to boil the solution will be higher. As a result, the boiling point of the solution will rise.

Elevation of boiling point is defined as an increase in boiling point $\Delta T_b = T_b - T_b^0$, where T_b^0 is the boiling point of pure solvent and T_b is the boiling point of solution.

Expression:

$$\Delta T_b = K_b m$$

The Boiling Point Elevation Constant, or K_b , is also known as the Molal Elevation Constant (Ebullioscopic Constant).

Calculation of the Solute's Molar Mass:

$$m = \frac{w_2 / M_2}{w_1 / 1000} = \frac{1000 \times w_2}{M_2 \times w_1}$$

By changing the molality value in the equation, we get

$$\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1}$$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1}$$

K_b When the molality of the solution is unity, it is defined as an increase in boiling point. The unit of K_b is kg mol^{-1} .

Determination of K_b : $K_b = \frac{R \times M_1 \times T_b^2}{1000 \times \Delta_{\text{vap}}}$

Where, R = gas constant (8.314 JK/mol)

T_f = freezing temperature in K M_1 = Molar mass of solvent in Kg/mol Δ_{vap} = enthalpy of vapourization in J/mol . The solution's vapour pressure curve is lower than that of pure water. ΔT_b denotes the elevation of a solvent's boiling point in solution, as seen in the diagram.

4.3. Depression in freezing point

A substance's freezing point is the temperature at which the vapour pressure of the substance in its liquid phase equals the vapour pressure of the substance in its solid phase.

When a non-volatile solid is introduced to a solvent, the vapour pressure drops until it equals that of a solid solvent at a lower temperature. As a result, the solvent's freezing point drops.

Depression in freezing point is defined as $\Delta T_f = T_f^0 - T_f$, where T_f^0 is the freezing point of pure solvent and T_f is the freezing point when a non-volatile solute is dissolved.

Expression

$$\Delta T_f = K_f m$$

Freezing Point Depression Constant, Molal Depression Constant, or Cryoscopic Constant are all names for K_f .

Calculation of molar mass of solute:

$$m = \frac{w_2 / M_2}{w_1 / 1000}$$

By Substituting This Molality Value into the Equation, We Get:

$$\Delta T_f = \frac{K_f \times w_2 / M_2}{w_1 / 1000}$$

$$\Delta T_f = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1}$$

$$M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$

When the molality of the solution is unity, it is defined as a decrease in freezing point.

K_f is measured in kg mol^{-1} .

Determination of K_f

$$K_f = \frac{R \times M_2 \times T_f^0}{1000 \times \Delta_{\text{fus}}}$$

Where

R = gas constant (8.314 JK/mol)

T_f = freezing temperature in K

M_1 = Molar mass of solvent in Kg/mol

$\Delta_{\text{fus}}H$ = enthalpy of fusion in J/mol T_f , or the depression of a solvent's freezing point in a solution, is depicted in this diagram.

4.4. Osmosis

When a semipermeable membrane separates a pure solvent and a solution, the solvent particles move through the membrane from the solvent side to the solution side. “Osmosis” is the name given to this phenomenon.

The semi-permeable membrane allows only small molecules to pass through, while blocking bigger solute molecules.

Osmotic Pressure: The excess pressure that must be given to a solution to prevent osmosis, or the flow of solvent molecules through a semipermeable barrier into the solution, is known as the osmotic pressure of a solution.

The osmotic pressure must be equal to the excess pressure. To prevent osmosis, a coating should be put on the solution.

Expression: At a given temperature T , osmotic pressure is proportional to the molarity, C , of a dilute solution.

Thus: $\pi = C R T$

The osmotic pressure is P , while the gas constant is R .

Calculation of Molar Mass:

$$\pi = (n_2 / V) RT$$

$$M_2 = \frac{w_2 RT}{\pi V}$$

Isotonic Solutions: Isotonic solutions are two liquids that have the same osmotic pressure at the same temperature.

In comparison to a more concentrated solution, a solution with a lower concentration or lower osmotic pressure is referred to as "Hypotonic."

In comparison to a dilute solution, a solution with a higher concentration or higher osmotic pressure is referred to as "hypertonic."

Reverse Osmosis: The solvent will flow from the solution into the pure solvent through the semi-permeable membrane if a pressure greater than the osmotic pressure is applied to the solution side.

The process is known as reverse osmosis.

Application: Desalination of seawater: Pure water is squeezed out of the seawater through the membrane when pressure greater than the osmotic pressure is applied.

4.5. Abnormal Molar Masses

The molecular mass of a substance is said to have aberrant molar mass when it is determined by investigating any of the colligative qualities and differs from the theoretically expected value.

Abnormal Molar Masses are Observed:

1. When the solute in the solution forms a connection.
2. When the solute in the solution undergoes dissociation.

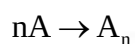
Van't Hoff Factor:

Van't Hoff proposed a factor [i] known as the van't Hoff Factor, to determine the level of association or dissociation.

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

$$i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}} = \frac{\text{Total no. of moles of particles after association(dissociation)}}{\text{No. of moles of particles before association(dissociation)}}$$

Association: the number of particles will always decrease because of connection, $i < 1$

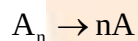


Let initial particles (n_i) = 1

$$\text{Final number} = 1 - \alpha + \alpha / n$$

$$\text{van't Hoff Factor } i = n_f / n_i = 1 - \alpha + \alpha / n$$

Dissociation: Due to dissociation, the number of particles will always rise, hence $i > 1$



Initial particle = 1

$$\text{Final particles} = 1 - \alpha + n\alpha, i = 1 - \alpha + n\alpha$$

Modified Expressions: Relatively lower solvent vapour pressure

$$\frac{p_1^0 - p_1}{p_1^0} = i \cdot \frac{n_2}{n_1}$$

$$\text{Elevation of boiling point } \Delta T_b = i K_b m$$

$$\text{Depression of Freezing point } \Delta T_f = i K_f m$$

$$\text{Osmotic pressure of solution, } \pi = i n_2 RT/V$$

5. Vapour Pressure

When the temperature of a liquid is raised, the molecules' desire to escape increases, and the vapour pressure rises.

In a liquid, the distribution of molecule kinetic energy.

Claussius - Clapeyron Equation

$$p_2 / p_1 = (\Delta H_{\text{VAP}} / R)(1/T_1 - 1/T_2)$$

Where ΔH_{VAP} represents the liquid's vaporisation enthalpy.

6. Thermodynamics of Dissolution

The process is exothermic if the contacts increase stronger during solution formation, and it is endothermic if they become weaker. In the dissolving process, ΔS is often positive. ΔG must be negative if the mixing process is spontaneous/natural.

6.1. Boiling Point Elevation

$$\Delta G_{\text{VAP}} = 0$$

$$T_b = \Delta H_{\text{VAP}} / \Delta S_{\text{VAP}}$$

The non-volatile solute improves the unpredictability of the solution phase while maintaining the same entropy of the vapours. As a result, ΔS_{VAP} lowers, causing the boiling point to rise.

The difference in their entropies of vapourization, ΔS_{VAP} , causes a solution's boiling point to be greater than that of a pure solvent.

6.2. Freezing Point Depression

$$\Delta G_{\text{FUS}} = 0$$

Due to the increase in the entropy of the solution, the entropy difference will increase in this scenario. According to the above relationship, an increase in entropy causes a drop in the freezing point.

A difference in their entropies of fusion, ΔS_{FUSION} , causes a solution to have a lower freezing point than a pure solvent.

7. Ostwald Walker Method

This is a common method for determining a solution's relative decrease in vapour pressure. Dry air is carried through three systems in order: solution, pure solvent, and finally a drying agent.

The decrease in weight of the vessels is represented by w_1 and w_2 , whereas the rise in weight of the third vessel is represented by w_3 . $w_1 \propto P_{\text{SOLUTION}}$ $w_2 \propto P_{\text{SOLVENT}} - P_{\text{SOLUTION}}$

(since the air was already saturated) $w_3 \propto P_{\text{SOLVENT}}$

The relative decrease in vapour pressure can be computed using the relationships.

A denotes the solvent and B represents the solute in the formulas below.

M_A = Molar mass of solvent

M_B = Molar mass of solute

W_A = Mass of solvent

V_B = volume of solute

V = volume of solution

d = density of solution

Some Important Formulas

$$\text{1. Mass of percentage (w/w)} = \frac{W_B}{W_A + W_B} \times 100$$

$$\text{Volume percentage(v/v)} = \frac{V_B}{V_A + V_B} \times 100$$

$$\text{Mass by volume percentage} \left[\frac{w}{v} \right] = \frac{W_B \times 100}{V(\text{mL})}$$

$$\text{Parts per million (ppm)} = \frac{W_B}{W_A + W_B} \times 10^6$$

$$\text{2. Mole Fraction of A, } x_A = \frac{n_A}{n_A + n_B}$$

$$\text{Mole fraction of B, } x_B = \frac{n_B}{n_A + n_B} \quad x_A + x_B$$

3. Molarity (M)

$$\text{Molarity(M)} = \frac{\text{Moles of solute}}{\text{Volume of solution in litre}} = \frac{n_B}{V_{(L)}} = \frac{W_B}{M_B \times V_{(L)}}$$

4. Molality (M)

$$\text{Molality(M)} = \frac{\text{Moles of solute}}{\text{Volume of solvent in kg}} = \frac{n_B}{W_A} \times 1000 = \frac{W_B \times 1000}{M_B \times W_A}$$

5. Normality(N)

$$\text{Normality(N)} = \frac{\text{Gram Equivalents of solute}}{\text{Volume of solution in litre}} = \frac{W_B}{\text{Gem of solute} \times V_{(L)}}$$

GEM= Gram Equivalent Mass

6. Relationship Between Molarity and Normality

A solution's normality (N) and molarity (M) are connected as follows:

$$\text{Normality} \times \text{Equivalent.mass(solute)} = \text{Molarity} \times \text{Molar mass(solute)}$$

7. Relationship Between Molarity and Mass percentage (p)

If p is the mass percentage of the solution and d is its density, then

$$\text{Molarity} = \frac{p \times d \times 10}{\text{Mol. mass(solute)}}$$

$$\text{Normality} = \frac{p \times d \times 10}{\text{Eq. mass(solute)}}$$

8. Dilution Formula

When a substance's solution is diluted by adding solvent from volume V_1 to volume V_2 , $M_1V_1 = M_2V_2$, similarly, $N_1V_1 = N_2V_2$.

9. Molarity of a Mixture

When V_1 mL of a molarity M_1 solution is mixed with another solution of the same substance with volume V_2 and molarity M_2 , the molarity of the resulting mixture of solution (M) can be calculated as follows:

$$M = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

10. Raoult's Law for Volatile Solute.

$p_A = p_A^0 x_A$ and $p_B = p_B^0 x_B$ The partial vapour pressures of component 'A' and component 'B' in the solution are p_A and p_B , respectively. The vapour pressures of pure components 'A' and 'B' are p_A^0 and p_B^0 , respectively.

Total vapour pressure $p = p_A + p_B = p_A^0 x_A + p_B^0 x_B$

11. Raoult's Law for Non-Volatile Solute

$$\frac{p_A^0 - p}{p_A^0} = x_B = \frac{n_B}{n_A + n_B} = \frac{n_B}{n_A} = \frac{W_B}{M_B} \times \frac{M_A}{W_A}$$

where x_B is the mole-fraction of the solute and $\frac{p_A^0 - p}{p_A^0}$ is the relative reduction in vapour pressure.

12. Elevation in Boiling Point.

$$\Delta T_b = K_b \times m$$

$$M_B = \frac{K_b W_B}{\Delta T_b \times W_A} \text{ Where } \Delta T_f = T_f^0 - T_f$$

14. Osmotic Pressure

$$(\pi) \pi = cRT$$

where 'c' is molarity

15. Van't Hoff Factor.

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

$$\text{Or } i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$$

$$\text{Or } i = \frac{\text{Total no. of moles of particles after association(dissociation)}}{\text{No. of moles of particles before association(dissociation)}}$$

Modified Forms of Colligative Properties:

$$(a) \frac{p_A^0 - p_A}{p_A^0} = i x_B$$

$$(b) \Delta T_b = i K_b m$$

$$(c) \Delta T_f = i K_f m$$

$$(d) \pi V = i n_B RT$$

16. $\alpha = \frac{i-1}{n-1}$ where α is the degree of dissociation, 'i' is the van't Hoff factor, and n is the number of ions produced per compound formula.

17. $\alpha = \frac{1-i}{1-\frac{1}{n}}$ where α is the degree of association, 1 is the number of molecules of solute that must associate to produce an associated molecule, and n is the number of molecules of solute that must associate to form an associated molecule where $\frac{1}{n} < 1$.

18. If $i > 1$, the solute will dissociate. If $i < 1$, the solute forms a bond with another substance. There is no linkage or dissociation when $i = 1$. The $i = \frac{1}{2}$ solute is a dimer.